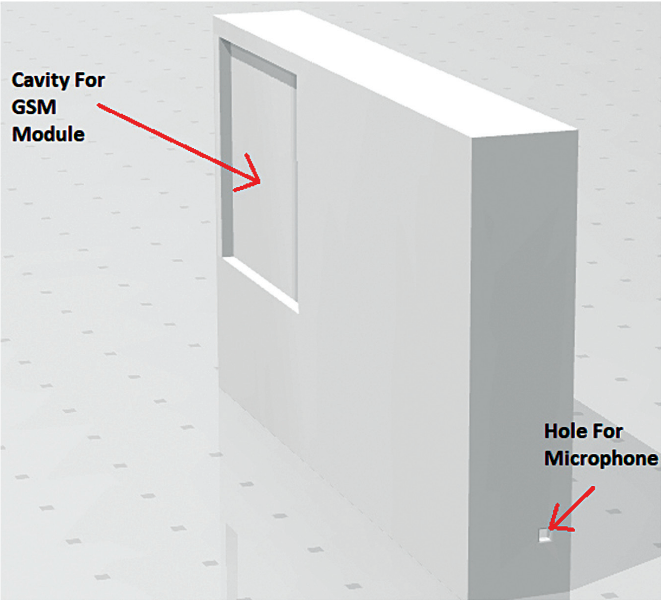

Fig. 4: Body design

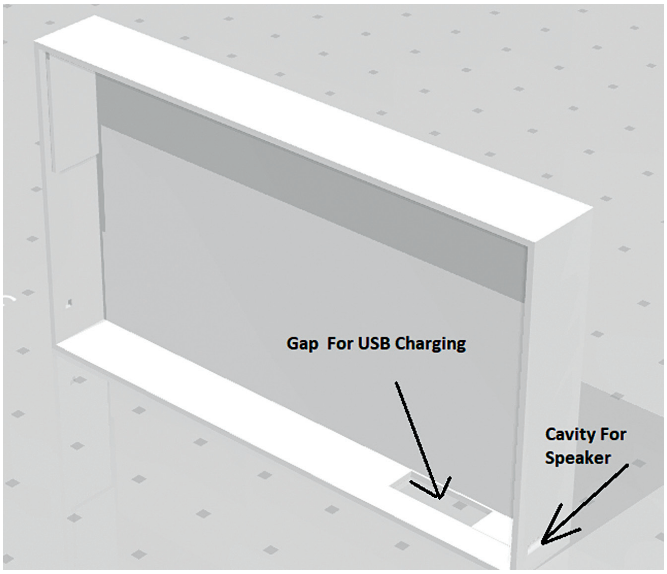

Fig. 5: Holes made in the phone's body

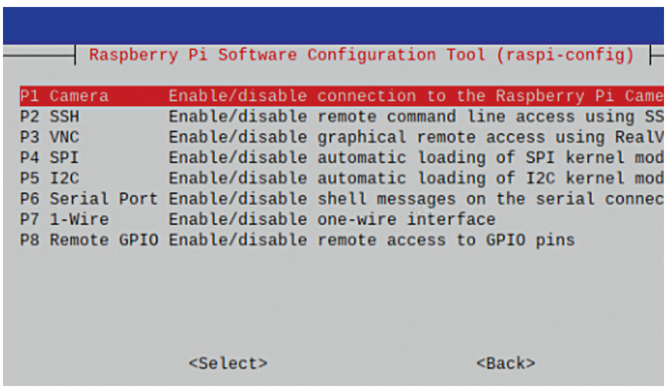

Fig. 6: Software configuration

MATERIAL REQUIRED

Component	Quantity	Description
Raspberry Pi Zero	1	Microcontroller
SIM800L	1	GSM module
Mini speaker	1	8-ohm,0.5W (mini)
Microphone	1	Micro sized mic
Battery and charger	1	3.3V rechargeable battery and charger
5.4cm E-ink touch display	1	Touch E-ink display
3D printed case	1	3D printed PLA case

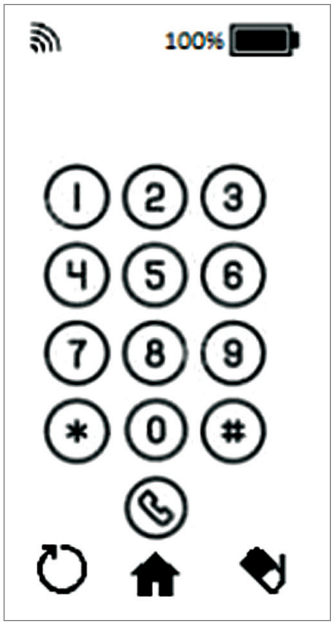

Fig. 7: UI for dialing screen

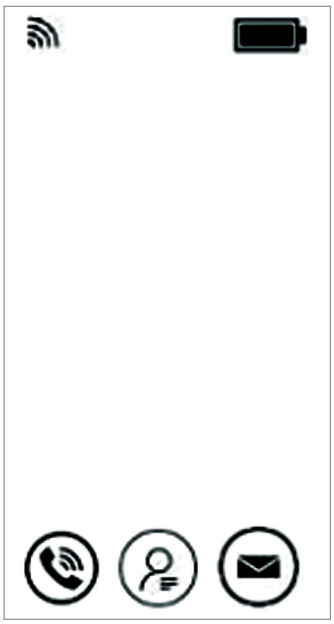

Fig. 8: UI for main screen

and, without any extra cost, you will get a full Linux based portable finger-sized computer as well! So, let us start collecting the required components, which are listed in the table above.

Designing

Though it is a totally home-made DIY phone, we can design it to look slim and attractive. So, for the phone, make its body the same size as the E-ink touch display. Then make space, as shown in Fig. 4, for embedding the GSM module inside the phone body.

Next make a tiny hole where the speaker is to be placed, so that you can hear the sound (see Fig. 5). Similarly, make holes for the mic and power input. Now, after designing the phone body, 3D print it.

Prepare the Raspberry Pi with its latest OS and then enable the SPI and I2C and serial ports of the Raspberry Pi. To do that, run the following command in